Ecological Complexity

A model of macroalgal decomposition

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Abstract:	Macroalgae are currently the focus of blue carbon, biological carbon dioxide (CO2) removal in the ocean. Carbon exported as seaweed detritus represents ~7.5% of anthropogenic CO2 emissions annually. Macroalgal blue carbon hinges on the fate of this detritus, so a reliable model of its decomposition is key. However, macroalgal decomposition has been neither consistently nor reliably modelled. Here I present the first model tailored to macroalgal decomposition. Informed by the theory of detrital photosynthesis, it treats detritus as an entity in limbo that can initially either grow or decompose. This is achieved by replacing the rate constant of the classic exponential decay model with a logistic function of time, mirroring the sigmoidal trend of detrital photosynthesis. The resulting mathematical model has three parameters that describe the initial relative growth rate, the time of photosynthetic death, and the final exponential decay rate which is comparable to the rate constant of the classic model. Using a Bayesian generative model with a beta prime likelihood built in the popular R and Stan languages, I show that the statistical implementation performs well given empirical data from a variety of contexts. The presented model can describe any macroalgal decomposition trajectory which enables formal comparison across studies for the first time. I anticipate that this will prove instrumental in resolving the debate around macroalgal CO2 removal.

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Dear editor,

I wish to submit a manuscript titled *A model of macroalgal decomposition* for consideration as a research article in *Ecological Complexity*. I confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

In this paper, I present the first model specifically tailored to macroalgal decomposition. This ecological process is notoriously difficult to model and no satisfactory solutions have been presented thus far. My model builds on the original exponential decay model first described in a highly cited paper on mathematical ecology 62 years ago but incorporates the recently developed theory of detrital photosynthesis according to which macroalgal detritus may behave completely contrary to the expectations of the classic model. I illustrate the efficacy of my model on a variety of published empirical datasets using a Bayesian implementation in R with Stan. I anticipate that my model will have far-reaching consequences for the assessment of blue carbon.

I believe my manuscript is suited to *Ecological Complexity* since it presents a novel mathematical model that describes an ecological process of broad interest. The model is embedded in theory and advances it by explicitly linking the ecophysiological theory of detrital photosynthesis to the ecological theory of decomposition for the first time. Besides showing the efficacy of this model using simulation, I turn it into practice using a Bayesian statistical model and empirical data. The resulting manuscript advances our understanding of the ecosystem function of marine carbon cycling and has socio-ecological implications for anthropogenic climate change mitigation. Thus the paper is of interest to your biogeochemical and theoretical ecological as well as your more applied readership.

All data and code are available in my open-access repository at github.com/lukaseamus/limbodeco and I encourage you to pass these on to the reviewers.

Sincerely,
Luka Seamus Wright

Ecological Complexity

A model of macroalgal decomposition

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Highlights

- First mathematical model of macroalgal decomposition
- Bayesian generative model with beta prime likelihood
- Examples from a variety of seaweeds including kelps
- Can describe any macroalgal decomposition trajectory
- Enables meta-analysis to infer macroalgal carbon cycle

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A model of macroalgal decomposition

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Abstract

Macroalgae are currently the focus of blue carbon, biological carbon dioxide (CO_2) removal in the ocean. Carbon exported as seaweed detritus represents ~7.5% of anthropogenic CO_2 emissions annually. Macroalgal blue carbon hinges on the fate of this detritus, so a reliable model of its decomposition is key. However, macroalgal decomposition has been neither consistently nor reliably modelled. Here I present the first model tailored to macroalgal decomposition. Informed by the theory of detrital photosynthesis, it treats detritus as an entity in limbo that can initially either grow or decompose. This is achieved by replacing the rate constant of the classic exponential decay model with a logistic function of time, mirroring the sigmoidal trend of detrital photosynthesis. The resulting mathematical model has three parameters that describe the initial relative growth rate, the time of photosynthetic death, and the final exponential decay rate which is comparable to the rate constant of the classic model. Using a Bayesian generative model with a beta prime likelihood built in the popular R and Stan languages, I show that the statistical implementation performs well given empirical data from a variety of contexts. The presented model can describe any macroalgal decomposition trajectory which enables formal comparison across studies for the first time. I anticipate that this will prove instrumental in resolving the debate around macroalgal CO_2 removal.

Keywords

Bayesian statistics, biogeochemistry, carbon cycle, degradation, kelp, ordinary differential equation

Introduction

In a seminal paper on mathematical ecology, Olson (1963) popularised the ordinary differential equation of leaf litter decomposition

$$\frac{dm}{dt} = -km \tag{1}$$

which describes the rate of change in litter mass m proportional to itself, a relationship which is parameterised by the decay constant k. This reflects our ecological understanding that given a theoretically limitless pool of decomposers, decomposition is proportional to the detrital pool. The equation integrates to

$$\ln\frac{m(t)}{m_0} = -kt \tag{2}$$

and simplifies to the familiar exponential decay function

$$m(t) = m_0e^{-kt} \tag{3}$$

where $m(t)$ is a function of the initial litter mass m_0, the decay constant k and time t. This function has prevailed as the model of choice for decomposition.

Interest in decomposition has been rekindled due to the hope that plants can mitigate anthropogenic climate change by fixing carbon dioxide (CO_2), assimilating it into carbohydrates, and sequestering it in their biomass. Carbon sequestration timescales of decades to centuries are being considered the minimum and ten millennia or longer are preferred (Babiker et al., 2022). Plant biomass can be split into two distinct carbon sinks: living plants and the detrital pool, of which the latter is considered the more promising. Often surprising to plant ecologists, plant detritus amounts to a biomass that is generally comparable and often far greater than that of living plants (Cebrián & Duarte, 1995, Wright & Kregting, 2023). But more importantly, living plants, even when considering stable or increasing populations, are limited by biomass-density constraints (Creed et al., 2019) and lifespans of typically less than a millennium (Thomas, 2013). Detritus on the other hand is believed to have the potential of cumulative sequestration over much longer timescales. Fossil fuels were created via the Carboniferous detrital pathway, mostly of scale trees

(Hibbett et al., 2016), and the hope of biological CO_2 removal essentially boils down to re-enacting fossil fuel generation.

Biological CO_2 removal in the ocean is referred to by the shorthand blue carbon (Babiker et al., 2022). Macroalgae have recently moved into the focus of blue carbon (Pessarrodona et al., 2023), primarily due to their substantial global area (Duarte et al., 2022), biomass packing (Creed et al., 2019), productivity (Pessarrodona et al., 2022) and detrital export (Pessarrodona et al., 2023, Filbee-Dexter et al., 2024). Notably, decomposition is disproportionally consequential for macroalgal blue carbon. Unlike their fellow blue carbon ecosystems—seagrass meadows, mangrove forests and salt marshes—seaweed forests cannot accumulate carbon in below-ground biomass and soil. There is thus a set limit for the living seaweed carbon sink which can be estimated as the product of 6.1 million km^2 global cover (Duarte et al., 2022), 1.2 ± 3 kg m^{-2} (mean ± standard deviation) standing biomass (Creed et al., 2019) and 28 ± 4.5 % carbon content (Pessarrodona et al., 2022). This yields 0.34 ± 0.87 kg C m^{-2} and 2.1 ± 5.3 Gt C globally, or only 19 ± 49 % of annual anthropogenic CO_2 emissions (Friedlingstein et al., 2025). However, seaweeds are thought to assimilate 0.32 ± 0.54 kg C m^{-2} y^{-1} (Pessarrodona et al., 2022) and 1.3 Gt C y^{-1} globally (Duarte et al., 2022) of which 61 ± 4.2 % is exported as detritus (Pessarrodona et al., 2023). Hence 0.19 ± 0.33 kg C m^{-2} y^{-1} and 0.81 ± 0.055 Gt C y^{-1} globally, or 7.5 ± 0.51 % of annual anthropogenic CO_2 emissions, may enter the detrital pool yearly. Note that these global standard deviations are underestimates because no uncertainty is reported by Duarte et al. (2022). Viewed cumulatively this is clearly the more substantial carbon sink. Decomposition alone therefore determines whether macroalgal blue carbon is significant or not, as is illustrated by the central role it takes in the latest macroalgal blue carbon models (Filbee-Dexter et al., 2024). Despite this, macroalgal decomposition remains comparatively poorly understood.

Phycologists have never been able to properly model macroalgal decomposition. Decomposition trajectories rarely match Equation 3 reasonably well (e.g. de Bettignies et al., 2020, Pedersen et al., 2021). Most studies have therefore either refrained from fitting a model (e.g. Birch et al., 1983, Hamersley et al., 2015) or most commonly fit linear models (e.g. Bedford & Moore, 1984, Brouwer, 1996, Frontier et al., 2021, Filbee-Dexter et al., 2022, Frontier et al., 2022, Wright et al., 2022). Meta-analyses of macroalgal k (Equation 3) are numerous (Enríquez et al., 1993, Banta et al., 2004, Cebrián & Lartigue, 2004, Filbee-Dexter et al., 2024, Kennedy & Blain, 2025, White & Norkko, 2025) but have generally assumed

Equation 3 by default and not assessed the validity of this model in the macroalgal context. It is evident that these meta-analyses are often hampered by the complexity and diversity of macroalgal decomposition trajectories (Kennedy & Blain, 2025, White & Norkko, 2025). The only serious attempts at modelling diverse trajectories have invariably invoked the logistic function alongside the exponential (Bourguès et al., 1996, Kennedy & Blain, 2025). This defeats the purpose of a model since logistic and exponential rate constants cannot be formally compared. Kennedy & Blain (2025) realised this when trying to compare half-lives ($\frac{\ln 2}{k}$ for Equation 3) across exponential and logistic models. While half-life for the latter is best estimated as the inflection point of the sigmoid, they ultimately resorted to forcing an exponential model to derive this parameter (Kennedy & Blain, 2025), which undermines fitting the logistic model in the first place.

We now know what causes the observed diversity in macroalgal decomposition trajectories. Early phycologists noticed the need to "pre-kill" macroalgal detritus (Birch et al., 1983, Smith & Foreman, 1984, Rieper-Kirchner, 1989, Brouwer, 1996) if it was to behave according to the expectations of the classic model (Equation 3). Recently it became evident that this is because excised macroalgal tissue can continue to photosynthesise for months (de Bettignies et al., 2020), which may allow macroalgal detritus to growth rather than decompose over the short term (Frontier et al., 2021, Pedersen et al., 2021, Filbee-Dexter et al., 2022, Frontier et al., 2022, Wright et al., 2022). To avoid semantic confusion, I use "detritus" for all detritus and "detritus *sensu stricto*" for dead detritus hereinafter. There is now extensive investigation into what has since been termed detrital photosynthesis (Frontier et al., 2021, Wright & Foggo, 2021, Frontier et al., 2022, Wright et al., 2022, Wright & Kregting, 2023, Wright et al., 2024) and macroalgae have been found to be unique among plants in having detritus that photosynthesises to an extent that is likely to influence its decomposition (Wright, in review). This knowledge enables the creation of a model that accurately describes macroalgal decomposition by taking into account the effect of detrital photosynthesis. It is probable that, ignorant of detrital photosynthesis, previous models have substantially underestimated macroalgal decomposition leading to overestimated blue carbon. To remedy this I here present a model with only three parameters that is tailored to the unique case of macroalgal decomposition. First I describe its mathematical form, then I implement it as a Bayesian statistical model in R (R Core Team, 2022) with Stan (Carpenter et al., 2017), and finally I validate the model against a variety of empirical examples.

The mathematical model

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… — Philip Pullman (1994) *The Tin Princess*

Macroalgal detritus is essentially in limbo. It may grow or decay, but not indefinitely. The goal is to model this temporary limbo. There are several possible approaches to describing this mathematically. As mentioned above, the logistic function has been the only attempt so far, but its rate constant is not comparable to that of Equation 3 and it cannot describe the full spectrum of possible decomposition trajectories (e.g. an increase followed by a decrease). Hence we must dismiss it. Another approach could be a composite function of a saturating function (e.g. rectangular hyperbola, hyperbolic tangent, exponential) and Equation 3. This may describe all possible trajectories but is somewhat of an artificial construct and more importantly would have more parameters than is practical. The best way forward is to stick with our trusty Equation 3 but rebuild it from its differential form (Equation 1). Instead of treating k as constant, we can let it be a function of time:

$$\frac{dm}{dt} = k(t)m \tag{4}$$

To my knowledge this approach has only been applied twice before, by Rovira & Rovira (2010) to describe the variety of terrestrial leaf litter decomposition trajectories and by Manojlovic (2021) to model seagrass decomposition (Trevathan-Tackett et al., 2020). Rovira & Rovira (2010) chose the form of $k(t)$ based on context while Manojlovic (2021) used a random walk, the simplest autoregression which defines k only in terms of k at the previous timepoint plus normal error:

$$k(t) \sim N(\, k(t - \Delta t)\,,\, \sigma\,) \tag{5}$$

where σ is the standard deviation of the normal distribution around the mean $k(t - \Delta t)$ and determines the k step size of the random walk. The timestep size Δt is typically set to 1. While this approach is the most flexible, it requires sophisticated Bayesian state space modelling with an ordinary differential equation solver. Manojlovic (2021) used LibBi

(Murray, 2015) and to my knowledge there is currently no way to implement this in the more popular Stan language (Manojlovic, personal communication, discourse.mc-stan.org/t/random-walk-in-rstan/34308). Furthermore, being the simplest type of order 1 autoregressive process, a random walk's only parameter is σ which has no inherent directionality and precludes informative prediction beyond the range of the data. Finally, no comparison can be made between studies beyond overlaying plots of $k(t)$, and formal comparison with studies that use the traditional model (Equation 3) is out of question. A model with informative parameters which can be formally compared is the way forward. Hence I here opt for the path laid out by Rovira & Rovira (2010).

Now the most important task is to find an appropriate function to describe k. Since the underlying process affecting k is detrital photosynthesis, it makes sense to start there. The change in detrital photosynthesis with time post excision, or detrital age, has mostly been described using various linear models (de Bettignies et al., 2020, Frontier et al., 2021, Wright & Foggo, 2021, Frontier et al., 2022, Wright et al., 2022, Wright & Kregting, 2023, Wright et al., 2024). There is no reasonable intrinsic mechanism causing photosynthesis to increase post excision. When an increase is observed it is likely due to an extrinsic factor such as a change in light environment (Frontier et al., 2021, Wright & Kregting, 2023). Instead, photosynthesis is mostly maintained near maximal levels until a breakpoint is reached and it decays to a minimum beyond which there logically is no further decline. Voilà, logistic decay! In Wright (in review) I have described detrital photosynthesis (p) as

$$p(t) = \frac{\alpha + \tau}{1 + e^{r(t - \mu)}} - \tau \qquad (6)$$

where α and τ may be read as the maximal CO_2 assimilation and production of the plant and its detritus respectively, r is the logistic rate of decay and μ is the time of photosynthetic death, more specifically the sigmoid inflection point. The naming of parameters is inspired by the story of the golem. As an entity in limbo, the golem is controlled by the word אמת (graecised τμα), symbolising life (א), transition (מ) and death (ת). The logistic function is optimal for modelling a transition between two alternate states and is thus the logical choice here.

In practice, I found the full four-parameter form of Equation 6 difficult to work with because r and μ are not identifiable and the model affords little control over the intercept, for which we known $p(0) \approx \alpha$ to be true. This usually requires a workaround, such as parameterising the model in terms of the logistic intercept ($r\mu$) or constraining the model with a joint prior on $\ln r + \ln \mu$ (Wright, in review). There are two ways to simplify Equation 6 to a three-parameter form: (1) set r constant and (2) describe r relative to μ, which is effectively the same as setting $r\mu$ constant. Reasonable values are $r = 1$ for option 1, which is roughly equivalent to photosynthetic death over one week, and $r = \frac{5}{\mu}$ for option 2 which implies that $r\mu = 5$, so $p(0) = 0.99(\alpha + \tau) - \tau \approx \alpha$. Theory would favour option 2 because we cannot assume a universal logistic rate of decay for detrital photosynthesis but we know $p(0) \approx \alpha$. Nonetheless, option 1 allows rapid photosynthetic decay independent of μ while option 2 implies that if detrital photosynthesis persists it must be due to slow decay initiated at $t = 0$. Accordingly the interpretation of α changes from detrital photosynthesis until photosynthetic death in option 1 to initial photosynthesis in option 2. While we know that the ultimate cause of photosynthetic decay is that initiating event (e.g. excision) and this agrees well with detrital photosynthesis trajectories (Wright, in review), data on macroalgal decomposition often seem to suggest that detrital photosynthesis is unperturbed until some more proximate mechanism causes it to fail (de Bettignies et al., 2020, Frontier et al., 2021, Frontier et al., 2022). I therefore tested both decomposition models on two empirical datasets (Brouwer, 1996, Frontier et al., 2022) given identical priors to compare performance. Option 2 has marginally better predictive performance according to leave-one-out cross-validation (Vehtari et al., 2017) but more importantly has smoother posteriors, while option 1 shows signs of bimodality, non-identifiability or jaggedness in posteriors (see github.com/lukaseamus/limbodeco for details). Thus I proceed with option 2 to describe k over time as

$$k(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t - \mu)}{\mu}}} - \tau \tag{7}$$

where α are the initial exponential growth rate (aka relative or specific growth rate) of detritus at detachment or excision, τ is the final exponential decay rate of detritus *sensu stricto* once detrital photosynthesis has ceased, and μ is the time of transition between the two states when the influence of detrital photosynthesis on decomposition ceases. The influence of each parameter on the trajectory of k is visualised in Figure 1. Inserting Equation 7 into Equation 4 yields

$$\frac{dm}{dt} = m(\frac{\alpha + \tau}{1 + e^{\frac{5(t - \mu)}{\mu}}} - \tau) \quad (8)$$

This equation could be used directly with statistical software that allows solving ordinary differential equations on the fly. But there is a simpler, generally applicable solution. Rovira & Rovira (2010) have shown that the logistic function can simply be integrated alongside the ordinary differential equation to yield a simpler model akin to the integrated form of Equation 3. Since we know our function is bounded by $t = 0$, we use an accumulation function. The definite integral of Equation 7 is

$$\int_0^t k(t)dt = t\alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}} \quad (9)$$

Thus when Equation 8 is integrated as a whole, we get the final form

$$m(t) = m_0 e^{t\alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}}} \quad (10)$$

Since decomposition data are best analysed as proportions of initial mass, $m_0 = 1$ which simplifies the function to one with only three parameters, the now familiar α, μ and τ. The influence of each parameter on the shape of the curve m describes over time is illustrated in Figure 1. Note that these parameters may be formally compared within and between studies, including those that used the traditional model (Equation 3). τ is the exponential decay rate of detritus *sensu stricto* which is not under the influence of detrital photosynthesis. Since the traditional model assumes that physiology does not play a role in detrital dynamics, k in Equation 3 also represents the exponential decay rate of detritus *sensu stricto*. Thus τ may be directly compared to k. In the next section I turn this mathematical model into a fully functional statistical model.

The statistical model

Maximum likelihood solutions to nonlinear modelling invariably demand the tedious task of guessing starting values. Rovira & Rovira (2010) followed this approach using the Levenberg-Marquardt algorithm which is more robust to poor starting values than the usual

Gauss-Newton algorithm. Despite this advantage, they still needed to manually approximate the starting values before passing them to the algorithm. Bayesian inference using Markov Chain Monte Carlo is more flexible in that there is no practical difference between fitting linear and nonlinear models (McElreath, 2019). Rather than starting values which can make or break the model, a prior probability distribution is provided, which at worst can lead to inefficient sampling and uncertain prediction. With the increase in computing power, every laptop can now run Markov Chain Monte Carlo. For these and other reasons Bayesian inference is quickly becoming the *status quo* in applied statistics across disciplines (McElreath, 2019). The most popular statistical language is R (R Core Team, 2022) and the most popular Bayesian language is Stan (Carpenter et al., 2017) which integrates nicely with R through various packages. Hence I developed the statistical model within this joint framework.

Several hierarchical functions are needed to describe a full statistical model. The one to rule them all is the likelihood function. It describes the probability distribution of the data. Most probability distributions are fully described by two parameters, usually the expected value and spread. For the normal distribution this is the mean and variance. Each of these likelihood parameters needs to be a function of the predictor. Finally the functions describing the likelihood parameters in turn have parameters, which in the Bayesian framework are expressed as probability distribution functions, the priors which once conditioned on the data turn into posteriors.

The mathematical model (Equation 10) describes the expected value of the likelihood. But what likelihood? We know that the data are proportions of initial mass that can exceed 1 due to growth caused by detrital photosynthesis. Several distributions are restricted to positive real numbers, such as the popular gamma distribution. But we can do better by thinking generatively. The proportions are ratios of mass to initial mass. Mass itself can only take positive real numbers and may be thought of as the sum of cells, growth increments etc. Thus the gamma distribution describes mass well. The ratio of two gamma distributions is a beta prime distribution. Technically this is only true if both gamma distributions share the same rate or scale, i.e. size of summand, and are independent. We can assume the former because both measures of mass can be broken down into the same components, e.g. cells. It is difficult to assume the latter because the fundamental assumption of the model is that the change in mass depends on mass (Equation 1, 3). Nonetheless, beta prime remains the best approximation of the true data distribution I can think of, so I use it as the likelihood here.

The beta prime distribution, typically parameterised by the two shape parameters α and β, is most intuitively parameterised in terms of mean $\mu = \frac{\alpha}{\beta - 1}$ and precision $\nu = \beta - 2$, where variance $\sigma^2 = \frac{\mu(1 + \mu)}{\nu}$ (Bourguignon et al., 2021). To reparametrise the likelihood, α and β need to be expressed in terms of μ and ν as $\alpha = \mu(1 + \nu)$ and $\beta = 2 + \nu$. We can either assume ν to be a constant (homoskedasticity) or a function of the predictor (heteroskedasticity). Given that for proportional data $m(0) = 1$ is predetermined, there is no uncertainty at $t = 0$, so ν is theoretically infinite to begin with. But with time comes uncertainty and we expect precision to decline. Sometimes remaining mass is calculated as a proportion of the mean rather than sample-specific initial mass so there is variability at $t = 0$ (e.g. Brouwer, 1996), but even then the variability at $t = 0$ is naturally lowest. Since $\nu > 0$ an appropriate model is

$$\nu(t) = (\epsilon - \theta)e^{-\lambda t} + \theta \tag{11}$$

where ϵ is ν at $t = 0$, λ is the rate of exponential decay in ν, and θ is ν as $t \rightarrow \infty$, inspired by its representation of Thánatos. Combining the beta prime likelihood and Equations 10 and 11 yields the full statistical model:

$$m \sim \text{Beta}'(\,\mu_m(1 + \nu)\,,\, 2 + \nu\,) \tag{12}$$

$$\mu_m = e^{t\alpha + \frac{\mu(\alpha + \tau)}{5}\ln\frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}}}$$

$$\nu = (\epsilon - \theta)e^{-\lambda t} + \theta$$

$$\alpha \sim N(\ldots),\ \text{Exp}(\ldots) \text{ or Gamma}(\ldots)$$

$$\mu, \tau, \epsilon, \lambda, \theta \sim \text{Exp}(\ldots) \text{ or Gamma}(\ldots)$$

where m is the proportion of remaining mass and μ_m is the mean of m, the remaining parameters being the same as in Equations 10 and 11. α may have a normal prior since it can take positive (initial growth) or negative (initial decomposition) values. However, in some cases it may be desirable to enforce initial growth with an exponential or gamma prior. All other parameters are restricted to positive values, so the exponential or gamma distribution is most suitable. I found generally suitable priors to be $\epsilon \sim$ Gamma(4 , 10^{-4}), $\lambda \sim$ Exp(1) and $\theta \sim$ Gamma(4 , 0.008), especially because these parameters are a bit abstract. I use these priors in all examples in the next section. For α, μ and τ I cannot recommend such generally applicable priors. As is usual with Bayesian workflow (McElreath, 2019), priors have to be chosen based on prior simulation. Furthermore, α, μ and τ are biologically meaningful and can thus be strongly informed by the literature. Some of the simplest examples that may be a reasonable starting point are $\alpha \sim N($ 0 , 0.01), $\mu \sim$ Exp(0.1) and $\tau \sim$ Exp(10). But in most cases these will have to be further constrained to ensure smooth sampling. The best next step would be to use a tighter normal prior for α, a gamma prior centred on half the range of t for μ and something along the lines of $\tau \sim$ Gamma(4 , 40) which has mean 0.1 and standard deviation 0.05. But in most cases you will be able to do far better.

Other likelihoods may also yield reasonable predictions if Equation 12 cannot be implemented in your statistical framework of choice. The gamma distribution may only get extreme values wrong due to its differently shaped tail and the normal distribution may still provide valid predictions of the mean even though it is not constrained to positive real numbers. If necessary, the model can also be simplified by assuming homoskedasticity, which may have minimal impact on mean predictions albeit inflating uncertainty near $t = 0$ and potentially underestimating uncertainty as $t \rightarrow \infty$.

There are some final considerations for the statistical model that are purely computational. Stan uses Hamiltonian Monte Carlo to explore probability space. Some of the expressions I have used in Equations 10 and 11 are easier to read but not friendly to the sampler. These can be improved for computational efficiency using the quotient and product rules of logarithms as

$$m(t) = e^{t\alpha + \frac{\mu(\alpha + \tau)}{5}\left(\ln(1 + e^{\frac{5(t - \mu)}{\mu}}) - \ln(1 + e^{-5}) \right)} \tag{13}$$

$$v(t) = e^{\ln(\epsilon - \theta) - \lambda t} + \theta \tag{14}$$

The $\ln(1 + e^{x})$ expressions in Equation 13 are further simplified in Stan using the wrapper function *log1p_exp* which is numerically stabler. An equivalent function doesn't exist in R but can be written as `log1p_exp <- function(x) ifelse( x > 0 , x +`

`log1p( exp(-x) ) , log1p( exp(x) ) )`. Note that Equation 11 performs better than Equation 14 in some cases. When prior variance for ϵ and/or θ is large which results in $\epsilon \leq \theta$ at initialisation, Equation 14 causes the log probability to evaluate to $-\infty$.

In terms of software, I use R v4.5.1 with the tidyverse package family v2.0.0 (Wickham et al., 2019) within the integrated development environment RStudio v2025.09.0+387 (RStudio Team, 2022) and run Stan with the R interface cmdstanr v0.9.0 (Gabry et al., 2024) via CmdStan v2.37.0 (Lee et al., 2017). All models have 8 Markov chains with 10^4 warmup and sampling iterations each. To assess quality, I use $\widehat{R}$ scores and visually scrutinise trace rank and pair plots with bayesplot v1.14.0 (Gabry et al., 2019). All reported results are posterior probabilities and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay, 2024). See github.com/lukaseamus/limbodeco for further details.

In the next section I will apply Equation 12 to four ecological effects in six published studies (Bourguès et al., 1996, Brouwer, 1996, Hamersley et al., 2015, de Bettignies et al., 2020, Frontier et al., 2021, Frontier et al., 2022). I either received data from the lead author (Frontier et al., 2021, Frontier et al., 2022) or digitised figures (Bourguès et al., 1996, Brouwer, 1996, Hamersley et al., 2015, de Bettignies et al., 2020) using WebPlotDigitizer v4.6 (Rohatgi, 2022). When several samples were collected at each timepoint, figures only yielded mean and standard deviation of remaining mass. The typical approach in such cases is measurement error modelling to account for the uncertainty of each mean so that means with larger standard deviations carry less weight in influencing the prediction (McElreath, 2019). However, at each timepoint de Bettignies et al. (2020) collected 5 samples, summarised as mean and standard deviation, for one treatment and only 1 sample for the other, resulting in a mixture of sample sizes and data levels. A measurement error model is not appliable in such a case. Furthermore, I believe latent variable models to be beyond the scope of this paper. I therefore simulated observations from summary statistics as random draws from the beta prime distribution in R using the *rbetapr* function from extraDistr v1.10.0 (Wolodzko, 2023) as

```
rbetapr(
  n ,
  mean * ( 1 + mean * ( 1 + mean ) / sd^2 ) ,
```
(15)

```
  2 + mean * ( 1 + mean ) / sd^2
)
```

where n, mean and sd are the reported sample size, sample mean and sample standard deviation. I derived this alternative parameterisation of the beta prime distribution by inserting $v = \frac{\mu(1+\mu)}{\sigma^2}$ into $\alpha = \mu(1 + v)$ and $\beta = 2 + v$ (Bourguignon et al., 2021).

Examples

Dead or alive

Brouwer (1996) provides a nice example of detrital photosynthesis influencing decomposition of the feathery brown seaweed *Desmarestia anceps* at Signy Island in Antarctica. She killed some of the detritus by heating to 50°C for 10–15 min prior to *in situ* decomposition (Brouwer, 1996). The resulting effect is remarkable but Brouwer (1996) was unable to quantify it due to lack of a suitable model at the time, so only provided an analysis of variance F statistic for change over time in both treatments and described the divergence in the decomposition trajectories.

The statistical model outlined in Equation 12 given this dataset and priors $\alpha \sim N(\ -0.005\ ,\ 0.003\ )$, $\mu \sim \text{Gamma}(\ \frac{200^2}{150^2}\ ,\ \frac{200}{150^2}\ )$ and $\tau \sim \text{Gamma}(\ \frac{0.1^2}{0.05^2}\ ,\ \frac{0.1}{0.05^2}\ )$ is visualised in Figure 2. Predictions across t for new observations of m based on the full likelihood ($\widetilde{m}$) are in the top panel, followed by the likelihood parameters v and μ_m, and k in the bottom panel. One could also overlay the probability distributions for the basal parameters α, μ and τ and ϵ and θ onto the plots of k and v respectively to present the full hierarchy of the model. To avoid clutter I have opted to only display μ, since this is the key parameter, marking the half-life of detrital photosynthesis, which, unlike α, τ, ϵ and θ, cannot be guessed by looking at predictions of k and v. Summaries of posterior probability distributions for the basal parameters describing μ_m and k are given in Table 1. I decided to pool τ, ϵ and θ across treatments. The experimental duration is not long enough to inform τ for the control treatment and it is reasonable to assume that untreated detritus decomposes at the same rate as pre-killed detritus once physiology has ceased. There is also no reason to suspect ϵ and θ to vary across treatments, since ϵ is only governed by the variability in initial experimental mass, which should be unaffected by treatment, and θ goes hand in hand with τ.

The model suggests that pre-killed *D. anceps* detritus decomposes much faster than the control (Figure 2). Untreated detritus has an initial relative growth rate of 0.037 ± 0.017 % d^{-1} (mean ± standard deviation, $P_{\alpha > 0} = 0.99$) while pre-killed detritus has an initial decay rate of 0.6 ± 0.29 % d^{-1} ($P_{\alpha < 0} = 0.98$) (Table 1). Detrital photosynthesis of *D. anceps* only ceases to influence decomposition after 778 ± 108 d (Table 1). Pre-killed detritus starts decomposing at the ultimate rate 750 ± 106 d earlier ($P_{\Delta\mu > 0} = 1$) (Table 1). Without physiology, detritus of *D. anceps* therefore decomposes in about 3.6 ± 0.56 % of the expected time.

Senescence

Hamersley et al. (2015) and de Bettignies et al. (2020) both studied the effect of tissue senescence on kelp decomposition. Hamersley et al. (2015) collected attached fresh and senescent fronds of the giant kelp *Macrocystis pyrifera* at Catalina Island in California along with naturally detached wrack and raft material which they regarded as detritus *sensu stricto*. de Bettignies et al. (2020) collected tissue from above and below the May cast abscission border of the kelp *Laminaria hyperborea* at Roscoff in Brittany. Their senescent blade thus represents tissue that the plant is in the process of abscising. Hamersley et al. (2015) refrained from fitting a model but de Bettignies et al. (2020) are among the few that fit Equation 3 to kelp decomposition data.

My models for *M. pyrifera* (Figure 3a) and *L. hyperborea* (Figure 3b) were both given priors $\alpha \sim N(\ 0\ ,\ 0.01\ )$ and $\mu \sim \text{Exp}(\ 0.1\ )$. But since de Bettignies et al. (2020) reported k values (Equation 3) of 0.0366 d^{-1} for senescent and 0.0107 d^{-1} for fresh lamina, I was able to better constrain the τ prior for this dataset. I therefore used $\tau \sim \text{Gamma}(\ \frac{0.2^2}{0.1^2}\ ,\ \frac{0.2}{0.1^2}\ )$ for *M. pyrifera* and $\tau \sim \text{Gamma}(\ \frac{0.0366^2}{0.02^2}\ ,\ \frac{0.0366}{0.02^2}\ )$ for *L. hyperborea*. For clarity I here only show predictions for k and μ_m with posterior probability distributions for μ, but the model provides the full set of information as in Figure 2. While species is constant across treatments in each study, the detritus has various natural origins rather than having been subjected to different experimental treatments as in the previous example. So I allowed all parameters except for ϵ to vary by treatment. I still pooled ϵ because, as is usual but unlike the previous example, there are no data to inform variability at $t = 0$ in these datasets.

Both models state that senescent fronds decompose faster than fresh ones, and detached ones decompose faster still (Figure 3). Initial decomposition rates for all treatments of *M. pyrifera*

(Table 1) are not distinguishable from zero ($P_{\alpha < 0}$ = 0.53–0.74) or from one another ($P_{\Delta\alpha > 0}$ = 0.53–0.58). However, the drop in *k* takes place 1.3 ± 0.81 d ($P_{\Delta\mu > 0}$ = 0.97) earlier for detached than senescent tissue for which in turn it happens 3.6 ± 1.3 d ($P_{\Delta\mu > 0}$ = 1) earlier than for fresh tissue (Figure 3a). The final decomposition rate for fresh tissue is 2.8 ± 1.2 % d^{-1} ($P_{\Delta\tau > 0}$ = 0.99) and 0.47 ± 1.2 % d^{-1} ($P_{\Delta\tau > 0}$ = 0.65) greater than for senescent and detached tissue. Fresh *L. hyperborea* has an initial growth rate of 1 ± 0.6 % d^{-1} ($P_{\alpha > 0}$ = 0.97) while senescent tissue of the same plant has an initial decay rate of 0.27 ± 1 % d^{-1} ($P_{\alpha < 0}$ = 0.6) (Table 1). Detrital photosynthesis in fresh tissue apparently ceases relatively early—after only 10 ± 3.1 d which is 6.7 ± 4.9 d ($P_{\Delta\mu > 0}$ = 0.93) after the senescent tissue—but its final decomposition rate is less than half that of senescent tissue ($\Delta\tau$ = 2 ± 0.38 % d^{-1}, $P_{\Delta\tau > 0}$ = 1) (Table 1). Due to this apparently relatively minor influence of detrital photosynthesis, decomposition rates are ultimately similar to those reported by (de Bettignies et al., 2020): 0.013 ± 0.0011 d^{-1} vs. 0.0107 d^{-1} for fresh and 0.032 ± 0.0036 vs. 0.0366 d^{-1} for senescent *L. hyperborea* (Table 1).

Light

The effect of depth on decomposition rates has repeatedly been evidenced for *Laminaria* species (Frontier et al., 2021, Frontier et al., 2022). Detritus of *L. hyperborea* and *L. ochroleuca* was either exposed to three simulated depths *ex situ* at Roscoff in Brittany (Frontier et al., 2021) or deployed at three depths *in situ* across the Channel at Plymouth in Devon (Frontier et al., 2022). Photophysiological measurements across detrital age have clarified that the amount of light available to detrital photosynthesis is the mechanism behind this effect (Frontier et al., 2021, Frontier et al., 2022). However, detrital photosynthesis and decomposition have not been quantitatively linked. Furthermore remaining mass was analysed using linear models which don't do a very good job given the nature of the data (Frontier et al., 2021, Frontier et al., 2022).

Given identical priors $\alpha \sim \text{Exp}(\ 100\ )$, $\mu \sim \text{Gamma}(\ \frac{50^2}{30^2}\ ,\ \frac{50}{30^2}\ )$ and $\tau \sim \text{Gamma}(\ \frac{0.1^2}{0.05^2}\ ,\ \frac{0.1}{0.05^2}\ )$, the results for both datasets are shown in Figure 4. Note that I am here making use of the possibility of constraining α to positive real numbers. All experimental tissue was excised from fresh parts of the lamina (Frontier et al., 2021, Frontier et al., 2022) so we expect initial growth. Due to the complexity of this dataset with three predictor variables, I only show $\widetilde{m}$, the posterior we would use to ultimately make predictions. In both cases I let α vary by species but not treatment, since there is no reasonable way for treatment to affect initial growth, and μ vary by species and treatment. As with α, for the first dataset I let τ vary by species but not by treatment following the rationale outlined above that detritus of the same origin can be expected to ultimately decompose at the same rate. For the second dataset I pooled τ across species and treatment since the experimental duration is not long enough to inform τ for *L. hyperborea* (Figure 4b). A more elegant solution would be partial pooling using multilevel modelling but that is beyond the scope of this paper. ϵ again was completely pooled and θ followed the pooling of τ for reasons outlined before.

Both models suggest that decomposition is faster at greater depths (Figure 4) for reasons outlined above. Decomposition also tends to be faster for *L. ochroleuca* than *L. hyperborea* (Figure 4), which is well established (Wright et al., 2022), but this does not hold true at greater depths (Figure 4a). Initial relative growth rates (Table 1) are not different between species for the first model ($P_{\Delta\alpha > 0} = 0.51$) but are 0.34 ± 0.13 % d^{-1} ($P_{\Delta\alpha > 0} = 1$) greater for *L. ochroleuca* than *L. hyperborea* in the second model. In the first model, detrital photosynthesis of *L. hyperborea* ceases 175 ± 38 d ($P_{\Delta\mu > 0} = 1$) later at the surface but 50 ± 17 d ($P_{\Delta\mu > 0} = 1$) and 12 ± 3.9 d ($P_{\Delta\mu > 0} = 1$) earlier at 15 and 30 m. Such a reversal in physiological persistence with depth is not evident in the second model, where *L. hyperborea* detritus photosynthesises 89 ± 35 d ($P_{\Delta\mu > 0} = 1$), 81 ± 32 d ($P_{\Delta\mu > 0} = 1$) and 27 ± 17 d ($P_{\Delta\mu > 0} = 1$) longer at 0.5, 1.5 and 3 m, probably due to the lesser depth range. In both models the maximal negative effect of depth on the longevity of detrital photosynthesis is stronger for *L. hyperborea* ($\Delta\mu$ = 87 ± 37 – 230 ± 38 d, $P_{\Delta\mu > 0} = 0.99$–1) than *L. ochroleuca* ($\Delta\mu$ = 25 ± 12 – 43 ± 6.4 d, $P_{\Delta\mu > 0} = 1$). So the detrital physiology of *L. hyperborea* is less tolerant to low light, which agrees with observed kelp forest zonation (Wright et al., 2022). The first model, with which it was possible to quantify differences in τ (Table 1), suggests that *L. ochroleuca* ultimately decomposes 0.6 ± 0.89 % d^{-1} ($P_{\Delta\tau > 0} = 0.75$) faster than *L. hyperborea*, independent of physiology. However, contrasting detrital physiology is clearly the main driver of the interspecific difference in decomposition timescales, as has been previously hinted at (Wright et al., 2022).

Season

Bourguès et al. (1996) provide perhaps the only data of macroalgal decomposition across all seasons. They studied decomposition of the ephemeral green seaweed *Ulvaria obscura* in the intertidal Arcachon Bay in France. Bourguès et al. (1996) collected fresh *U. obscura* but buried it below 1 cm of sediment where anoxia apparently persisted and there clearly was no light for detrital photosynthesis. Given these experimental conditions, it would be surprising to see decomposition affected by physiology. Yet Bourguès et al. (1996) were faced with the challenge of having to model decomposition trajectories of sigmoidal appearance for which they chose the logistic function. Excised macroalgae can survive exceptional periods of darkness (Wright et al., 2024) so it may be that saprotrophs were initially inhibited by chemical defence despite the absence of photosynthesis. Even if this was not the case because anoxia immediately killed the seaweed, and the model therefore is conceptually wrong, it is still useful because it avoids having to fit fundamentally incomparable models as Bourguès et al. (1996) have done.

Given this dataset and priors $\alpha \sim \mathrm{Exp}(\ 100\ )$, $\mu \sim \mathrm{Gamma}(\ \frac{25^2}{10^2}\ ,\ \frac{25}{10^2}\ )$ and $\tau \sim \mathrm{Gamma}(\ \frac{0.1^2}{0.05^2}\ ,\ \frac{0.1}{0.05^2}\ )$, $\widetilde{m}$ is visualised in Figure 5. Again I make use of constraining α to positive values because the seaweed was initially alive. Season, which for all intents and purposes is temperature, affects detrital physiology as well as other aspects that may drive decomposition post-physiology such as saprotroph metabolic rates (Filbee-Dexter et al., 2022, Wright et al., 2024). So all parameters are expected to be affected by season. I therefore only pooled ϵ across treatments for reasons outlined above.

According to the model decomposition is generally faster in warmer seasons (Figure 5). This agrees well with the known effect of temperature on decomposition (Filbee-Dexter et al., 2022, Wright et al., 2024). Initial growth rate is not strikingly different between any two seasons ($P_{\Delta\alpha > 0}$ = 0.54–79), perhaps unsurprisingly given data sparsity immediately after the start of the experiment (Figure 5). Differences are more pronounced in the decline of k over time. k shifts at a similar time in autumn and winter ($P_{\Delta\mu > 0}$ = 0.52) but does so 9 ± 3.6 d ($P_{\Delta\mu > 0}$ = 1) and 19 ± 3.3 d ($P_{\Delta\mu > 0}$ = 1) earlier in spring and summer than in autumn (Table 1). This may be the detrimental effect of increased temperature on detrital physiology (Wright et al., 2024) since spring and summer are generally warmer than autumn and winter, unless we believe the experimental conditions immediately killed the seaweed. The final value of k is 6 ± 6.4 % d^{-1} ($P_{\Delta\tau > 0}$ = 0.84), 2.8 ± 5.5 % d^{-1} ($P_{\Delta\tau > 0}$ = 0.73) and 5 ± 6 % d^{-1} ($P_{\Delta\tau > 0}$ = 0.81) lower—i.e. decomposition is faster and τ is greater—in spring, summer and autumn than in winter (Table 1). This is the effect of seasonal temperature differences on factors other than detrital physiology. Again the physiological effect, albeit putative in this case, is more pronounced, as has been suggested in the context of temperature (Wright et al., 2024).

Conclusion

Macroalgal decomposition has neither been consistently nor reliably modelled. But a realistic model is the key to understanding macroalgal blue carbon. I have presented the first decomposition model tailored to macroalgae by taking into account the theory of detrital photosynthesis. This is made possible by treating the exponential decay rate as a function of time rather than a constant, a function which is informed by the latest understanding of detrital photosynthesis. My model only has three parameters which describe the initial relative growth rate (α), the half-life of detrital photosynthesis (μ), and the final exponential decay rate (τ) which is directly comparable to that of the classic model (k). I have shown that these parameters are successfully estimated and meaningful for empirical data from a variety of contexts. Given a set of parameter estimates, the model can reliably predict new observations, even beyond the duration of the experiment. Although I have written the model in R with Stan, it can be implemented in any statistical framework that allows robust nonlinear modelling. Failing to account for detrital photosynthesis almost invariably leads to underestimation of macroalgal decomposition. I therefore expect that this model will increase estimates of macroalgal decomposition and thus have major implications for macroalgal blue carbon accounting. I will be using this model to clarify the role of kelp forests in CO_2 removal and hope others find it useful too.

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I declare no conflicts of interest. Importantly I am not funded by any private enterprise with an interest in selling carbon credits.

Data and code availability

Data and code to reproduce this paper and apply my model to your own data are available at github.com/lukaseamus/limbodeco. Please cite this paper as well as the model (Wright, 2025).

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Tables

Table 1. Macroalgal decomposition parameter estimates (mean ± standard deviation) for Equation 10 (μ_m in Equation 12) based on 8×10^4 posterior samples. Units are given in parentheses. For readability the exponential rates α and τ are given as daily percentages, but they are modelled as daily proportions. Pooled estimates within a study are marked with the same symbol.

Species	**Treatment**	**α (% d^{-1})**	**μ (d)**	**τ (% d^{-1})**
Brouwer 1996				
Desmarestia anceps	Control	0.037 ± 0.017	778 ± 108	7.2 ± 0.91*
	Pre-killed	−0.6 ± 0.29	28 ± 4.1	7.2 ± 0.91*
Hamersley *et al.* 2015				
Macrocystis pyrifera	Fresh	−0.31 ± 0.55	5.9 ± 1	12 ± 1.1
	Senescent	−0.19 ± 1	2.3 ± 0.75	9.3 ± 0.54
	Detached	−0.076 ± 1	0.94 ± 0.3	12 ± 0.47
de Bettignies *et al.* 2020				
Laminaria hyperborea	Fresh	1 ± 0.6	10 ± 3.1	1.3 ± 0.11
	Senescent	−0.27 ± 1	3.3 ± 3.7	3.2 ± 0.36
Frontier *et al.* 2021				
Laminaria hyperborea	0 m	0.03 ± 0.017*	259 ± 48	4.1 ± 0.49*
	15 m	0.03 ± 0.017*	27 ± 9.2	4.1 ± 0.49*
	30 m	0.03 ± 0.017*	1.9 ± 1.2	4.1 ± 0.49*
Laminaria ochroleuca	0 m	0.037 ± 0.032†	57 ± 6.5	4.6 ± 0.71†
	15 m	0.037 ± 0.032†	80 ± 18	4.6 ± 0.71†
	30 m	0.037 ± 0.032†	14 ± 3.7	4.6 ± 0.71†
Frontier *et al.* 2022				
Laminaria hyperborea	0.5 m	0.24 ± 0.042*	105 ± 22	6.1 ± 2.2*
	1.5 m	0.24 ± 0.042*	94 ± 20	6.1 ± 2.2*
	3 m	0.24 ± 0.042*	42 ± 12	6.1 ± 2.2*
Laminaria ochroleuca	0.5 m	0.58 ± 0.13†	43 ± 9.2	6.1 ± 2.2*
	1.5 m	0.58 ± 0.13†	35 ± 8.7	6.1 ± 2.2*
	3 m	0.58 ± 0.13†	21 ± 3.8	6.1 ± 2.2*
Bourguès *et al.* 1996				
Ulvaria obscura	Spring	0.43 ± 0.4	14 ± 2.2	15 ± 4.1
	Summer	0.7 ± 0.71	4.4 ± 1.8	12 ± 2.5
	Autumn	0.6 ± 0.33	23 ± 2.8	14 ± 3.4
	Winter	0.29 ± 0.3	24 ± 7.3	9.5 ± 4.9

Figures

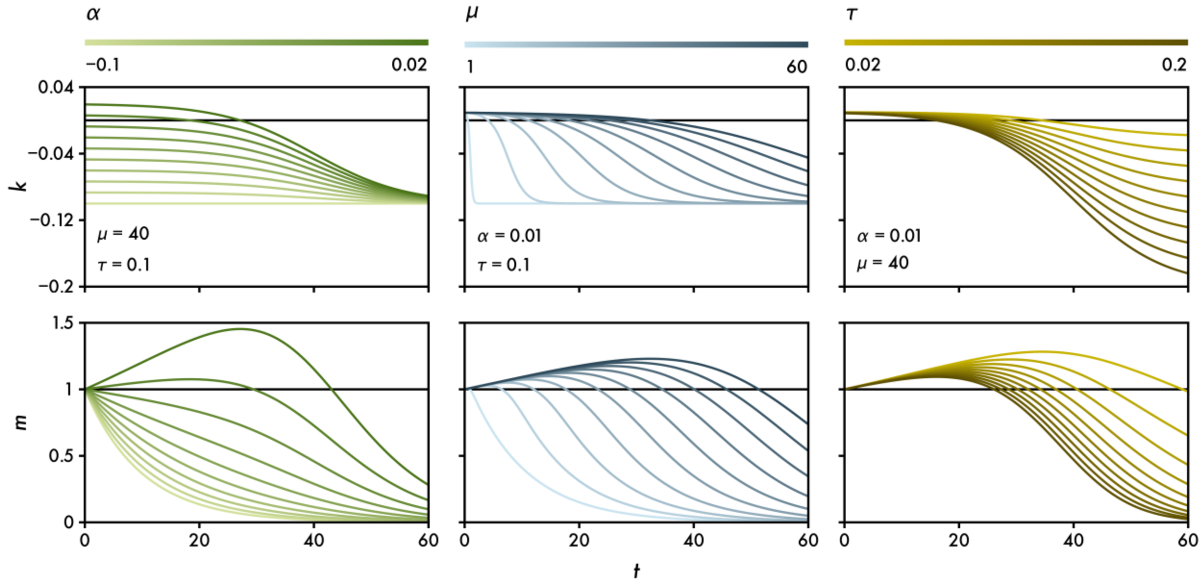


Figure 1. The mathematical model. Some reasonable parameter values for α, μ and τ and how they independently determine the behaviour of the exponential decay rate k (Equation 7) and the proportion of remaining mass m (Equation 10) over time t.

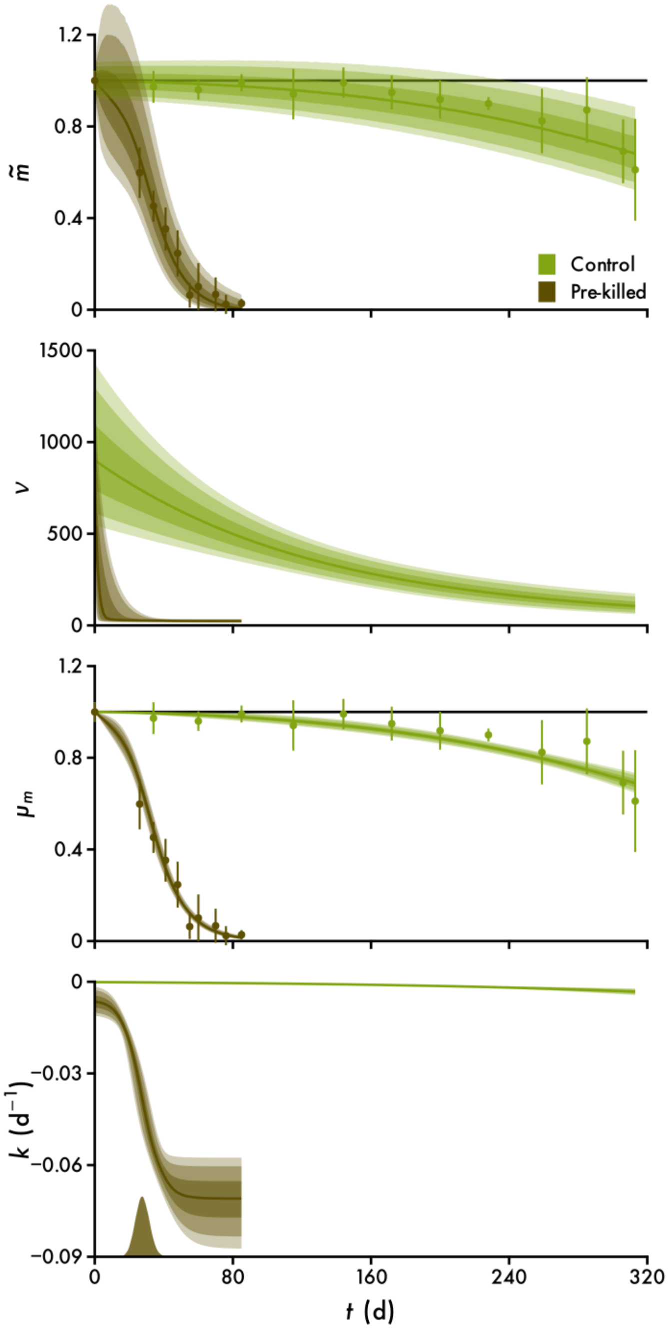


Figure 2. The statistical model. Example 1: the effect of physiology on the decomposition of *Desmarestia anceps*, shown as the proportion of remaining mass over time t (data from Brouwer, 1996). Data are plotted alongside predictions of new observations ($\tilde{m}$), the precision (ν), the mean (μ_m) and the exponential decay rate (k) (Equation 12). Points with error bars denote mean ± standard deviation of 5 samples. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

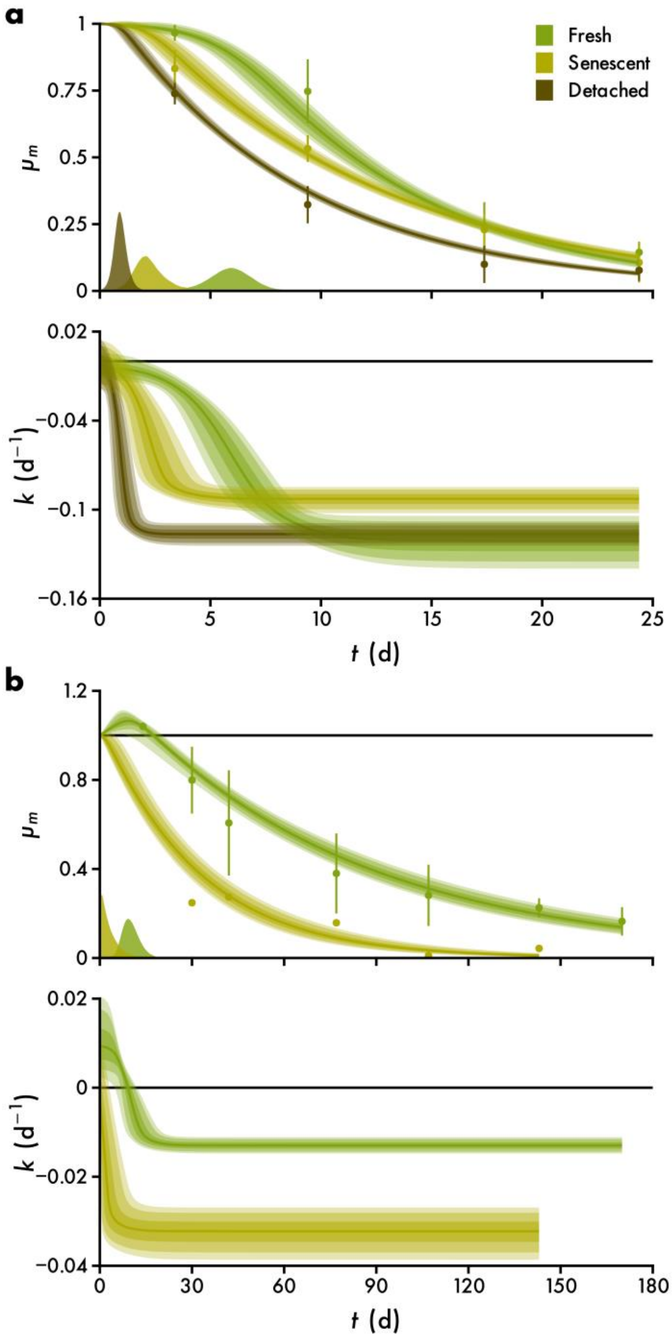


Figure 3. The statistical model. Example 2: the effect of senescence on the decomposition of *Macrocystis pyrifera* (**a**, data from Hamersley et al., 2015) and *Laminaria hyperborea* (**b**, data from de Bettignies et al., 2020), shown as the proportion of remaining mass over time t. Data are plotted alongside predictions of the mean (μ_m) and the exponential decay rate (k) (Equation 12). Points with error bars denote mean ± standard deviation of 5 samples. Points without error bars are 1 sample. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

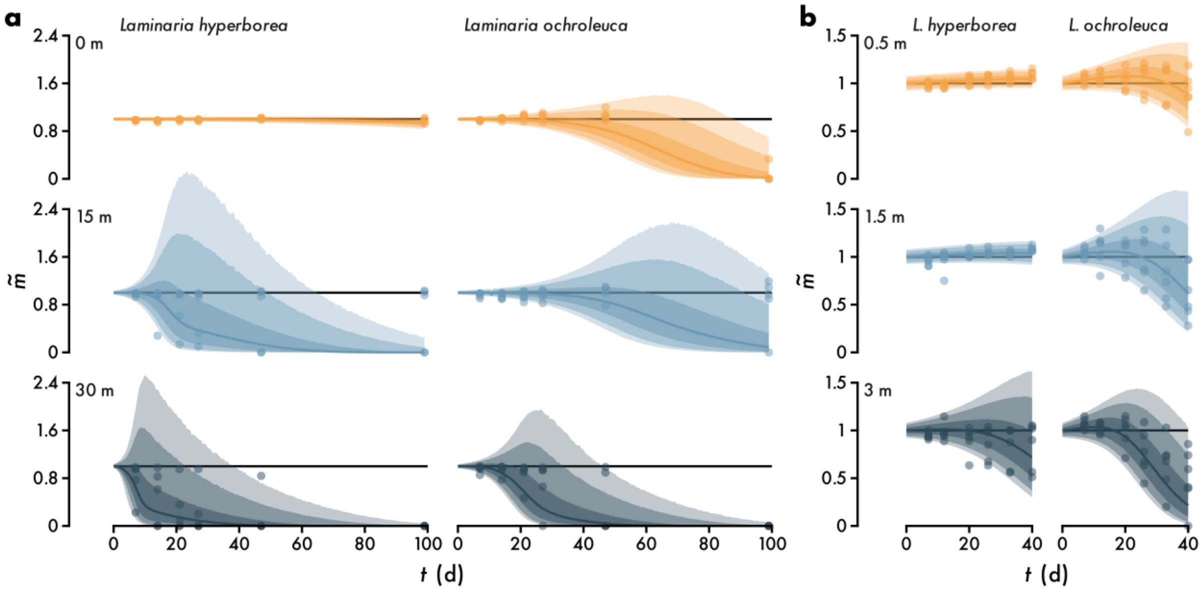


Figure 4. The statistical model. Example 3: the effect of light attenuation with depth on the decomposition of *Laminaria hyperborea* and *Laminaria ochroleuca ex situ* (**a**, data from Frontier et al., 2021) and *in situ* (**b**, data from Frontier et al., 2022), shown as the proportion of remaining mass over time t. Data are plotted alongside predictions of new observations ($\tilde{m}$, Equation 12). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

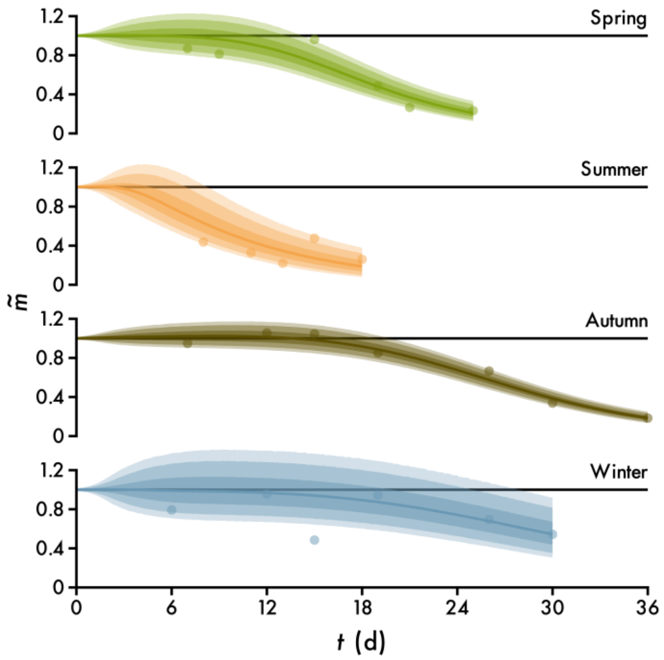
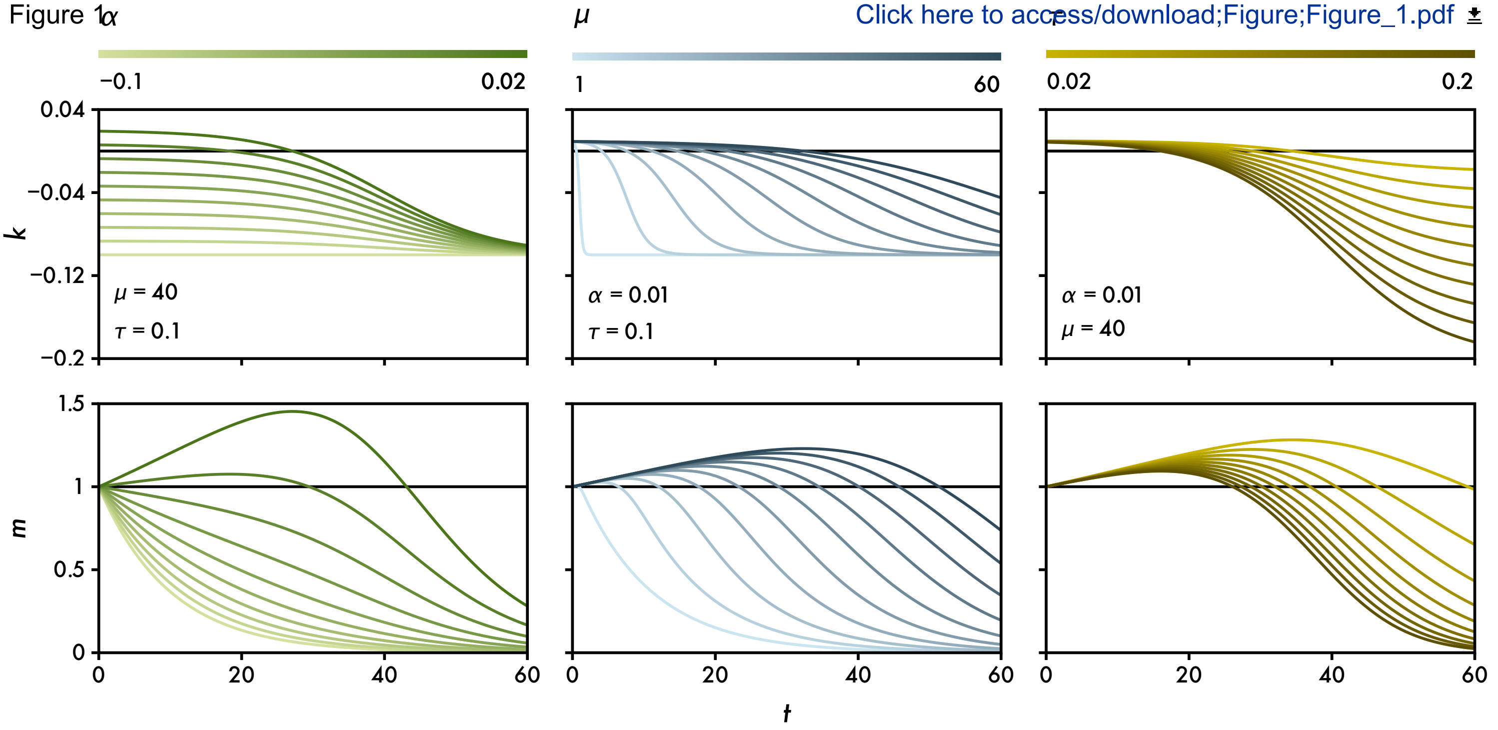


Figure 5. The statistical model. Example 4: the effect of season on the decomposition of *Ulvaria obscura* (data from Bourguès et al., 1996). Data are plotted alongside predictions of new observations ($\tilde{m}$, Equation 12). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

Figure 1



α
−0.1
0.02
μ
1
60
0.02
0.2
0.04
−0.04
−0.12
−0.2
k
μ = 40
τ = 0.1
α = 0.01
τ = 0.1
α = 0.01
μ = 40
1.5
1
0.5
0
m
0
20
40
60
t

Figure 2

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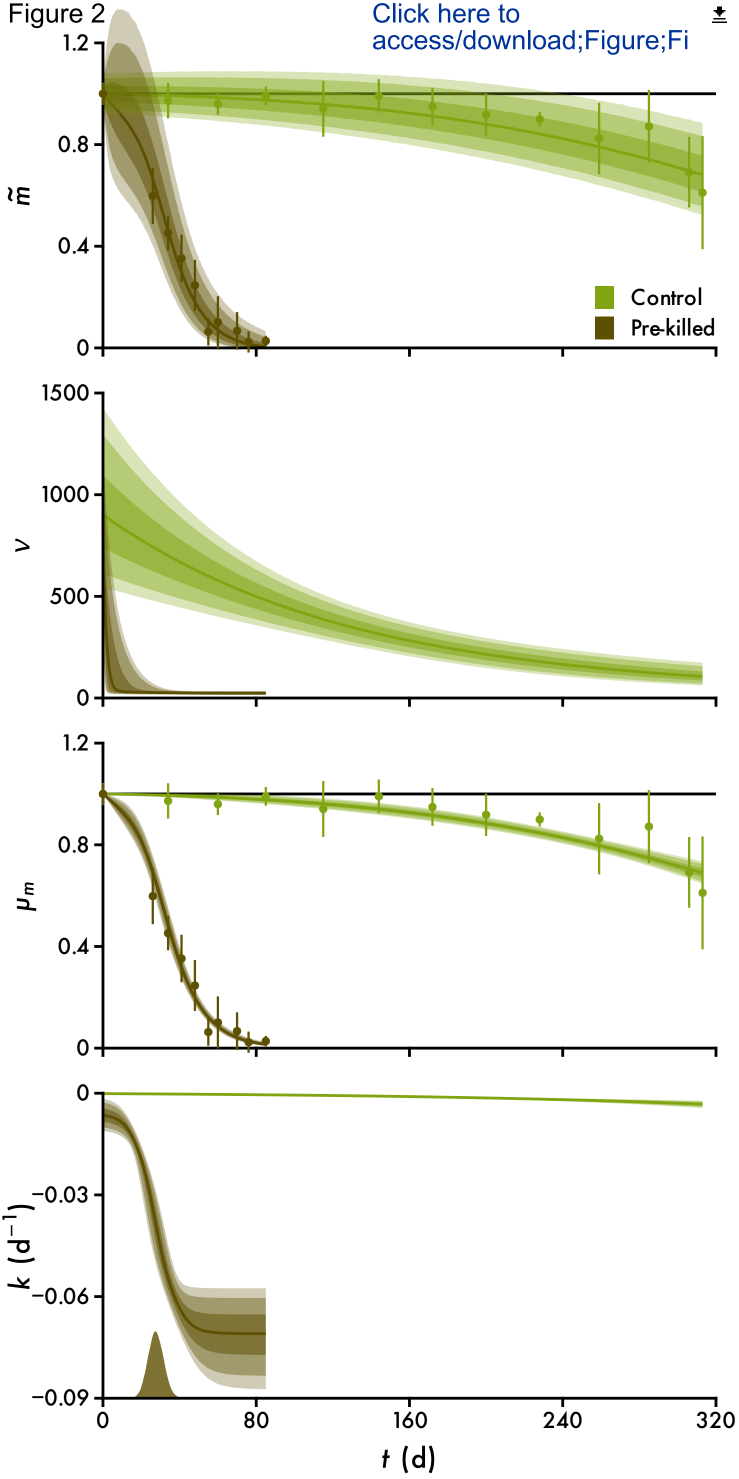

Figure 3

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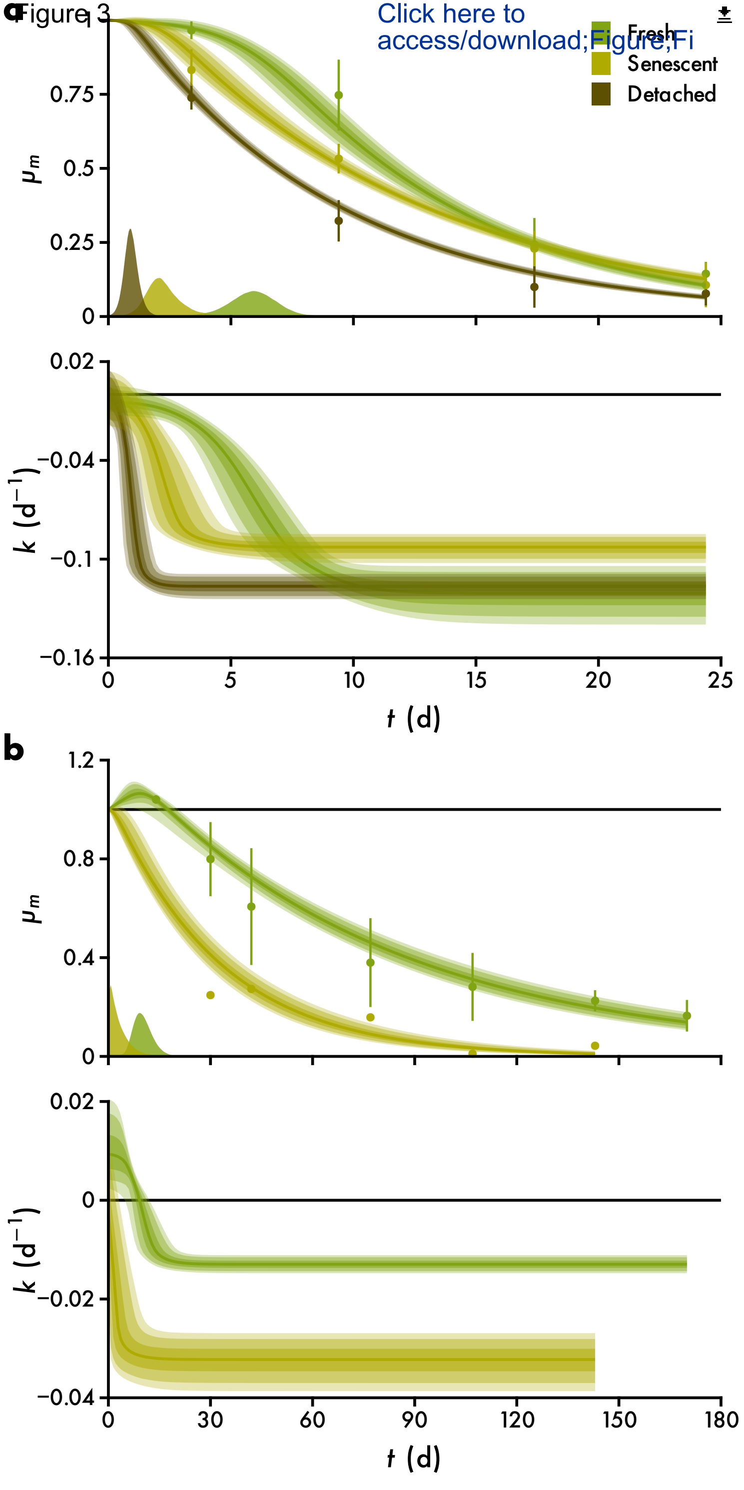

Figure 4

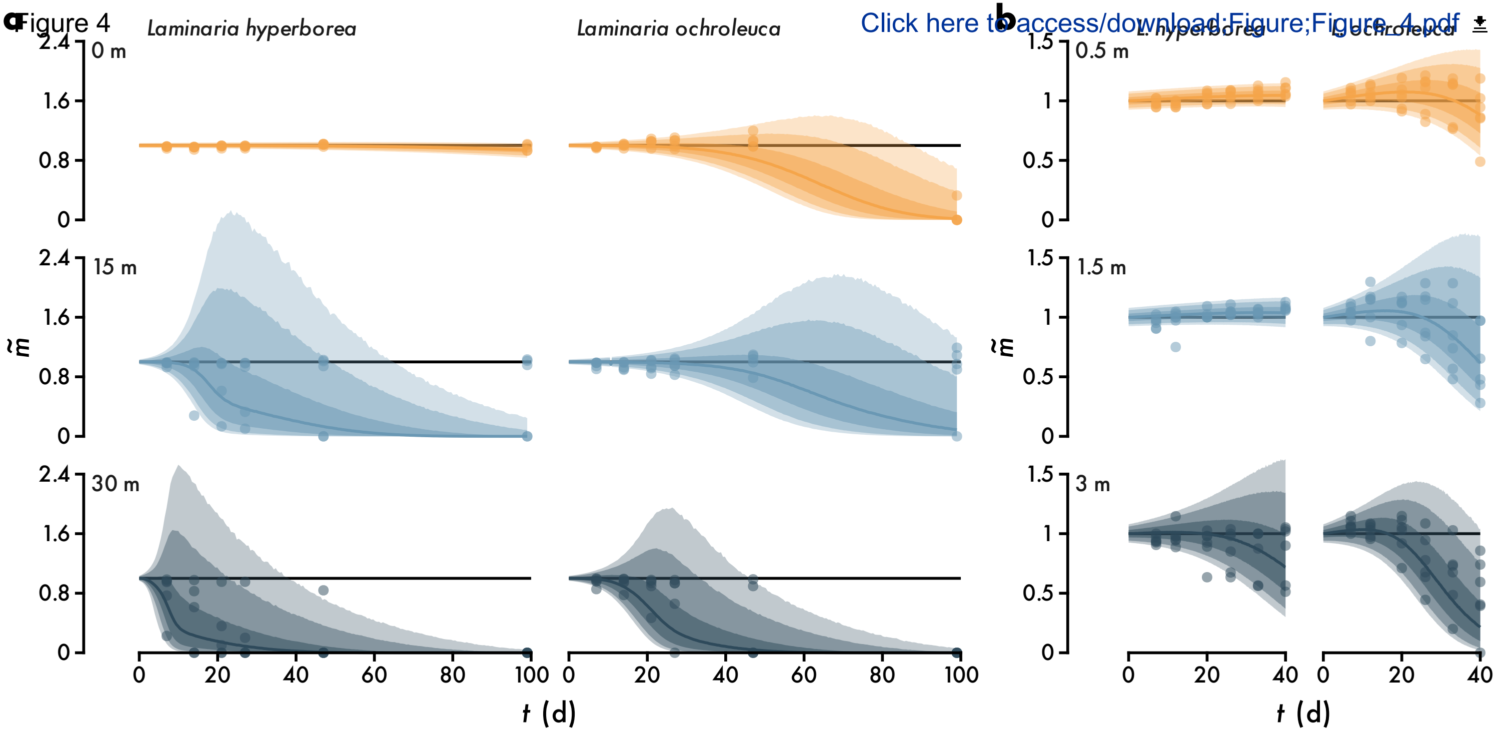
a
Laminaria hyperborea
Laminaria ochroleuca
0 m
15 m
30 m
2.4
1.6
0.8
0
m̃
0
20
40
60
80
100
t (d)
b
L. hyperborea
L. ochroleuca
0.5 m
1.5 m
3 m
1.5
1
0.5
0
m̃
0
20
40
t (d)

Figure 5

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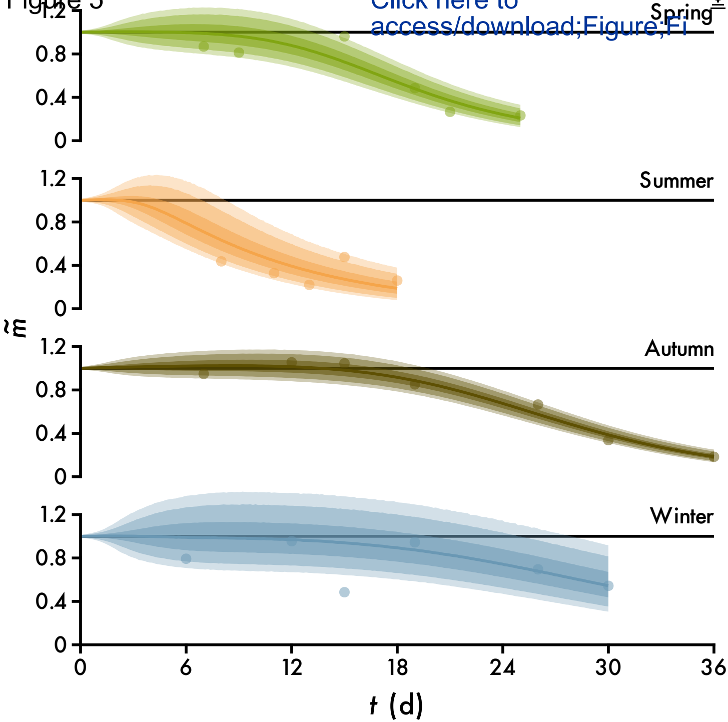

Ecological Modelling

A model of macroalgal decomposition

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Declaration of competing interest

I declare no conflicts of interest. Importantly I am not funded by any private enterprise with an interest in selling carbon credits.